

Axcel[®]

VALACYCLOVIR- 500

TABLET

COMPOSITION:

Each tablet contains
Valacyclovir Hydrochloride
equivalent to Valacyclovir

500mg

PRESENTATION:

White to off white, film-coated, oblong tablet.

INDICATIONS:

Axcel Valacyclovir-500 is indicated for the treatment of herpes zoster (shingles). Axcel Valacyclovir-500 accelerates the resolution of pain; it reduces the duration, and the proportion, of patients with zoster-associated pain, which includes acute and post-herpetic neuralgia. Axcel Valacyclovir-500 is indicated for the treatment of herpes simplex infections of the skin and mucous membranes, including initial and recurrent genital herpes. Axcel Valacyclovir-500 can prevent lesion development when taken at the first signs and symptoms of an HSV recurrence. Axcel Valacyclovir-500 is indicated for the prevention (suppression) of recurrent herpes simplex infections of the skin and mucous membranes, including genital herpes. Axcel Valacyclovir-500 can reduce transmission of genital herpes when taken as suppressive therapy and combined with safer sex practices (particularly the use of condoms).

PHARMACOLOGY:

Mechanism of Action:

Valacyclovir is a prodrug (L-valine ester) of the antiviral acyclovir. Valacyclovir is rapidly and almost completely converted in man to acyclovir and valine, probably by the enzyme referred to as valacyclovir hydrolase. Acyclovir is active against herpes simplex virus (HSV) type 1 and type 2 and also against varicella-zoster virus (VZV), cytomegalovirus (CMV), Epstein-Barr Virus (EBV) and human herpes virus 6 (HHV-6). The phosphorylation process or intracellular conversion activity of acyclovir from monophosphate with subsequent conversion by cellular enzyme (viral thymidine kinase (TK)) to the cellular diphosphate and the active triphosphate. The viral DNA synthesis and viral replication is halted by inhibiting the herpes virus DNA polymerase enzyme. Only infected cell is selected in this process. Studies in animal and *in vitro* have found various sensitivities and these virus are inhibited by concentration of acyclovir which is readily achieved clinically.

The most susceptible is herpes simplex virus (HSV) type 1, and then type 2 followed by varicella-zoster virus (VZV). The Epstein-Barr Virus (EBV) and cytomegalovirus (CMV) also susceptible to acyclovir to a lesser extent. But cytomegalovirus (CMV) does not activate by thymidine kinase and may act via a different mechanism. Epstein-Barr Virus (EBV) may reduce thymidine kinase activity but its DNA polymerase is very sensitive to inhibition by acyclovir triphosphate. Extensive monitoring of clinical HSV and VZV isolates from patients receiving acyclovir therapy or prophylaxis has revealed that virus with reduced sensitivity to acyclovir is extremely rare in the immunocompetent and is only found infrequently in severely immunocompromised individuals e.g. organ or bone marrow transplant recipients, patients receiving chemotherapy for malignant disease and people infected with the human immunodeficiency virus (HIV). Resistance is normally due to a thymidine kinase (TK) deficient phenotype with results in a virus which is profoundly disadvantaged in the natural host. Infrequently, reduced sensitivity to acyclovir has been described as a result of subtle alterations in either the virus thymidine kinase or DNA polymerase. The virulence of these variants resembles that of the wild-type virus.

Pharmacokinetics:

General characteristics:

Valacyclovir is well absorbed after oral administration and rapidly converted to acyclovir and valine. This conversion is probably mediated by enzyme isolated from human liver referred to as valacyclovir hydrolase. Bioavailability of acyclovir after valacyclovir administration is reported to be 54%, and is not reduced by food. Mean peak acyclovir concentrations are 10 – 37 micron (2.2 – 8.3 mcg/ml) after single doses of 250 -2000mg valacyclovir to healthy subjects with normal renal function, and occur at a median time of 1 – 2 hours post dose. Peak plasma concentrations of valacyclovir are only 4% of acyclovir levels, occur at a median time of 30 to 100 minutes post dose, and are at or below the limit of quantification 3 hours after dosing. The valacyclovir and acyclovir pharmacokinetic profiles are similar after single and repeat dosing. Binding of valacyclovir to plasma proteins is very low (15%). In patients with normal renal function the plasma elimination half-life of acyclovir after both single and multiple dosing with valacyclovir is approximately 3 hours. In patients with end-stage renal disease, the average elimination half-life of acyclovir after valacyclovir administration is approximately 14 hours. Less than 1% of the administered dose of valacyclovir is recovered in the urine as unchanged drug. Valacyclovir is eliminated in the urine as acyclovir (greater than 80% of the recovered dose) and the known acyclovir metabolite, 9-(carboxymethoxy)methylguanine (CMMG).

Characteristics in patients:

Herpes zoster and herpes simplex do not significantly alter the pharmacokinetics of valacyclovir and acyclovir after oral administration of Axcel Valacyclovir - 500. In a pharmacokinetic study of Valacyclovir and Acyclovir during late pregnancy, the steady-state daily acyclovir AUC (area under the curve) following valacyclovir 1000mg was approximately 2 times greater than that observed with oral acyclovir at 1200mg daily. In patients with HIV infection,

the disposition and pharmacokinetics characteristics of acyclovir after oral administration of single or multiple doses of 1000mg or 2000mg valacyclovir are unaltered compared with healthy subjects. In transplant recipients receiving valacyclovir 2000mg 4 times daily, acyclovir peak concentrations are similar to or greater than those in healthy volunteers receiving the same dose. The estimated daily AUCs are appreciably greater.

DOSAGE AND ADMINISTRATION:

Treatment of herpes zoster:

Adult : 1g, three times daily for 7 days.

Treatment of herpes simplex infections:

Adult : 500mg twice daily;

for recurrent episode: treatment should be for 5 days; for initial episodes, which can be more severe, treatment may have to be extended to 10 days. Dosing should begin as early as possible. For recurrent episodes of herpes simplex, this should ideally be during the prodromal period or immediately after the first signs or symptoms appear.

Suppression (Prevention) of recurrences of herpes simplex infections:

In immunocompetent adult patients: 500mg to be taken once daily. Some patients with very frequent recurrences (e.g. 10 or more per year) may gain additional benefit from the daily dose of 500mg being taken as a divided dose (250mg, twice daily).

For immunocompromised adult patients: 500mg twice daily.

Reduction of transmission of genital herpes:

In immunocompetent heterosexual adult patients with 9 or fewer recurrences per year, 500mg daily should be taken by infected partner in order to reduce transmission to a sexual partner negative for HSV-2 antibodies. Safer sex practices (particularly condom use) should be maintained, and sexual contact avoided if lesions are present. There are no data on the reduction of transmission beyond 8 months in other patient populations.

Dosage in renal impairment:

Dosage may need to be reduced in patients with renal impairment, as shown in the table below :

Therapeutic Indication	Creatinine Clearance (ml/min)	Axcel Valacyclovir-500 Dosage
Herpes zoster	15-30	1g twice a day
Patients on haemodialysis	Less than 15 -	1g once a day 1g daily after haemodialysis
Herpes simplex (treatment)	15-30	No modification
Patients on haemodialysis	Less than 15 -	500mg once a day 500mg daily after haemodialysis

Therapeutic Indication	Creatinine Clearance (ml/min)	Axcel Valacyclovir-500 Dosage
Suppression (prevention) of herpes simplex a) immunocompetent patients - Patients on haemodialysis	Less than 15 -	250mg once a day 250mg daily after haemodialysis
b) immunocompromised patients - Patients on haemodialysis	Less than 15 -	500mg once a day 500mg daily after haemodialysis

Dosage in hepatic impairment:

Studies with a 1g unit dose of valacyclovir show that dose modification is not required in patients with mild or moderate cirrhosis (hepatic synthetic function maintained). Pharmacokinetic data in patients with advanced cirrhosis, (impaired hepatic synthetic function and evidence of portal-system shunting) do not indicate the need for dosage adjustment, however, clinical experience is limited. For higher doses (4g or more/day), please refer to Warning & Precaution.

Dosage in children:

There are no data available on the use of valacyclovir in children.

Dosage in elderly:

Dosage modification is not required unless renal function is significantly impaired (see Dosage in renal impairment above). Adequate hydration should be maintained.

CONTRAINDICATION:

Contraindicated in patients known to be hypersensitive to valacyclovir, or acyclovir.

WARNING AND PRECAUTION:

Axcel Valacyclovir-500 should be used with caution in patients with renal impairment and dose should be adjusted according to creatinine clearance. (see Dosage and Administration)

Hydration Status: Care should be taken to ensure adequate fluid intake in patients who are at risk of dehydration, particularly the elderly.

Use in Renal Impairment: The Axcel Valacyclovir-500 dose should be adjusted in patients with renal impairment. Patients with a history of renal impairment are also at increased risk of developing neurological effects. (see Side Effects)

Use of Higher Doses in Hepatic Impairment and Liver Transplantation: There are no data available on the use of high doses of valacyclovir (≥4 g/day) in patients with liver disease. Caution should therefore be exercised when administering higher doses of Axcel Valacyclovir-500 to these patients.

Use in Genital Herpes: Suppressive therapy with Axcel Valacyclovir-500 reduces the risk of transmitting genital herpes. It does not cure genital herpes or completely eliminate the risk of transmission. In addition to therapy with Axcel Valacyclovir-500, it is recommended that patients

use safer sex practices (particularly the use of condoms).

Effects on the Ability to Drive or Operate Machinery: The clinical status of the patient and the adverse event profile of valacyclovir should be borne in mind when considering the patient's ability to drive or operate machinery. There have been no studies to investigate the effect of Axcel Valacyclovir-500 on driving performance or the ability to operate machinery. Further, a detrimental effect on such activities cannot be predicted from the pharmacology of valacyclovir.

Preclinical Safety Data:

Carcinogenicity: Valacyclovir was not carcinogenic in bio-assays performed in mice and rats.

Mutagenicity: The results of mutagenicity tests *in vitro* and *in vivo* indicate that valacyclovir is unlikely to pose a genetic risk to humans.

Immune: Drug reaction with eosinophilia and systemic symptoms (DRESS), which can be life-threatening or fatal, has been reported in association with valacyclovir treatment. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. If signs and symptoms suggestive of DRESS appear, valacyclovir should be withdrawn immediately and an alternative treatment considered (as appropriate). If a patient has developed DRESS with the use of valacyclovir, treatment with valacyclovir must not be restarted in this patient at any time.

PREGNANCY AND LACTATION:

Teratogenicity: Valacyclovir was not teratogenic in rats or rabbits. Valacyclovir is almost completely metabolized to acyclovir. Subcutaneous administration of acyclovir in internationally accepted tests did not produce teratogenic effects in rats or rabbits. In additional studies in rats, foetal abnormalities were observed at subcutaneous doses that produce plasma levels of 100mcg/ml and maternal toxicity.

Fertility: Valacyclovir did not affect fertility in male or female rats dosed by the oral route.

Use in pregnancy: There are limited data on the use of Axcel Valacyclovir - 500 in pregnancy. Axcel Valacyclovir - 500 should only be used in pregnancy if the potential benefits of treatment outweigh the potential risk.

Use in lactation: Acyclovir, the principle metabolite of valacyclovir is excreted in breast milk. Following oral administration of a 500mg dose of valacyclovir, peak acyclovir concentrations (C_{max}) in breast milk ranged from 0.5-2.3 (median 1.4) times the corresponding maternal acyclovir serum concentrations. The median acyclovir concentration in breast milk was 2.24mcg/mL (9.95 μ M). With a maternal valacyclovir dosage of 500mg twice daily, this level would expose a nursing infant to a daily oral acyclovir dosage of about 0.61mg/kg/day. The elimination half-life of acyclovir from breast milk was similar to that for serum. Unchanged valacyclovir was not detected in maternal serum, breast milk or infant urine. Caution is advised in Axcel Valacyclovir-500 is to be administered to a nursing woman. However, acyclovir, the active metabolite of valacyclovir is used to treat neonatal herpes simplex at IV doses of 30mg/kg/day.

SIDE EFFECTS:

Adverse reactions are listed as follows by MedDRA body system organ class and by frequency:
The frequency categories used are:
Very common (≥ 1 in 10);
Common (≥ 1 in 100 and < 1 in 10);
Uncommon (≥ 1 in 1000 and < 1 in 100);
Rare (≥ 1 in 10,000 and < 1 in 1000);
Very rare (< 1 in 10,000).

Clinical trial data have been used to assign frequency categories to adverse reactions if, in the trials, there was evidence of an association with valacyclovir (ie, there was a statistically significant difference between the incidence in patients taking valacyclovir and placebo). For all other adverse events, spontaneous post-marketing data has been used as a basis for allocating frequency.

Clinical Trial Data:

Nervous System Disorders:

Common: Headache.

Gastrointestinal Disorders:

Common: Nausea.

Post-Marketing Data:

Blood and Lymphatic System Disorders:

Very Rare: Leukopenia, thrombocytopenia. Leukopenia is mainly reported in immunocompromised patients.

Immune System Disorders:

Very Rare: Anaphylaxis.

Drug reaction with eosinophilia and systemic symptoms (DRESS)

Psychiatric and Nervous System Disorders:

Rare: Dizziness, confusion, hallucinations, decreased consciousness.

Very Rare: Agitation, tremor, ataxia, dysarthria, psychotic symptoms, convulsions, encephalopathy, coma.

The above events are reversible and usually seen in patients with renal impairment or with other predisposing factors.

Respiratory, Thoracic and Mediastinal Disorders:

Uncommon: Dyspnoea.

Gastrointestinal Disorders:

Rare: Abdominal discomfort, vomiting, diarrhoea.

Hepatobiliary Disorders:

Very Rare: Reversible increases in liver function tests. These are occasionally described as hepatitis.

Skin and Subcutaneous Tissue Disorders:

Uncommon: Rashes, including photosensitivity.

Rare: Pruritus.

Very Rare: Urticaria, angioedema.

Renal and Urinary Disorders:

Rare: Renal impairment.

Very Rare: Acute renal failure.

Others: There have been reports of renal insufficiency, microangiopathic haemolytic anaemia and thrombocytopenia (sometimes in combination) in severely immunocompromised patients, particularly those with advanced HIV disease, receiving high doses (8g daily) of valacyclovir for prolonged periods in clinical trials. These findings have been observed in patients not treated with valacyclovir who have the same underlying or concurrent conditions.

DRUG INTERACTIONS:

No clinically significant interactions have been identified. Acyclovir is eliminated primarily unchanged in the urine via active renal tubular secretion. Any drug administered concurrently that competes with this mechanism may increase acyclovir plasma concentrations following Axcel Valacyclovir-500 administration. Following 1g Axcel Valacyclovir-500, cimetidine and probenecid increase the AUC of acyclovir by this mechanism, and reduce acyclovir renal clearance. However, no dosage adjustment is necessary at this dose because of the wide therapeutic index of acyclovir. In patients receiving higher doses of Axcel Valacyclovir-500 (≥ 4 g/day), caution is required during concurrent administration with drugs which compete with acyclovir for elimination, because of the potential for increased plasma levels of one or both drugs or their metabolites. Increases in plasma AUCs of acyclovir and of the inactive metabolite of mycophenolate mofetil, an immunosuppressant agent used in transplant patients, have been shown when the drugs are co-administered. Care is also required (with monitoring for changes in renal function) if administering high-dose Axcel Valacyclovir - 500 (≥ 4 g/day) with drugs which affect other aspects of renal physiology (eg, cyclosporin, tacrolimus).

OVERDOSAGE & TREATMENT:

There are at present limited data available on overdosage with Axcel Valacyclovir-500. However patients have ingested single overdoses of up to 20g of acyclovir, which is only partially absorbed in the gastrointestinal tract, usually without toxic effects. Accidental, repeated overdoses of oral acyclovir over several days have been associated with gastrointestinal effects (such as nausea and vomiting) and neurological effects (headache and confusion). Overdosage of intravenous acyclovir has resulted in elevations of serum creatinine and subsequent renal failure. Neurological effects including confusion, hallucinations, agitation, seizures and coma have been described in association with intravenous overdosage.

Management:

Patients should be observed closely for signs of toxicity. Haemodialysis significantly enhances the removal of acyclovir from the blood and may, therefore, be considered at management option in the event of symptomatic overdose.

STORAGE:

Store below 30°C. Protect from light.

KEEP OUT OF REACH OF CHILDREN
JAUIH DARI KANAK-KANAK

PACK QUANTITIES:

Available in blister pack of 10 x 10's.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder & Manufactured By:



Kotra Pharma (M) Sdn. Bhd.
No. 1, 2 & 3, Jalan TTC 12, (90082-V)
Cheng Industrial Estate,
75250 Melaka, Malaysia.